Artin's Rational Singularities and The Fundamental Cycle

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Abstract. This study note establishes the formal algebraic and combinatorial framework of Artin's rational surface singularities and their corresponding fundamental cycles, providing the prerequisite machinery for studying the planar contact structures analyzed by Plamanevskaya and Starkston. We first review the rigorous definition of the fundamental cycle Z as the unique minimal anti-nef effective divisor supported on the exceptional locus of a minimal good resolution. We detail Artin's constructive algorithm, highlighting how the negative-definiteness of the intersection matrix guarantees finite convergence to Z. We then examine Artins characterization theorem, exploring how the analytic vanishing of the first higher direct image sheaf $R^1\pi_*\mathcal{O}_{\tilde{X}}$ reduces purely to the combinatorial vanishing of the arithmetic genus $p_a(Z) = 0$. Finally, we specialize to the geometric conditions under which the fundamental cycle is strictly reduced, translating these numerical criteria into explicit valence bounds on plumbing graphs that dictate the topology of Stein fillings.

Contents

1. The Set–up: Resolution of a Surface Singularity

Recall the following from previous notes. Let (X, x_0) be a normal complex surface singularity, viewed as the germ of a two-dimensional complex analytic variety with an isolated singularity at $x_0 \in X$. By normality, x_0 is completely isolated, and the local ring $\mathcal{O}_{X,x_0}$ is an integrally closed Noetherian domain.

Definition 1.1 (Good Resolution). A **good resolution** of the singularity (X, x_0) is a proper biholomorphic map

$$\pi : \tilde{X} \longrightarrow X$$

from a smooth complex analytic surface $\tilde{X}$ such that:

1. The restriction $\pi|_{\tilde{X}\setminus\pi^{-1}(x_0)} : \tilde{X} \setminus \pi^{-1}(x_0) \longrightarrow X \setminus \{x_0\}$ is an isomorphism.
2. The **exceptional locus** $E := \pi^{-1}(x_0)$ is a connected compact connected subvariety of $\tilde{X}$ decomposes into a finite union of irreducible projective curves:
$$E = \bigcup_{i=1}^{n} E_i$$
3. Each E_i is a smooth, non-singular complex curve of genus $g(E_i) = h^1(E_i, \mathcal{O}_{E_i})$.
4. The components intersect **transversely** and at most pairwise. That is, $E_i \cap E_j$ is either empty or consists of a finite number of points where the curves cross analytically transversely, and $E_i \cap E_j \cap E_k = \emptyset$ for distinct indices i, j, k.

Theorem 1.2 (Grauert's Contractibility Criterion). *Let $\tilde{X}$ be a smooth complex surface, and $E = \bigcup_{i=1}^{n} E_i$ a connected configuration of compact irreducible curves. The exceptional locus E can be analytically contracted to an isolated normal singularity if and only if the **intersection matrix** $M = (E_i \cdot E_j)_{1\leq i,j\leq n}$ is strictly **negative-definite**.*

1.1. The Divisor Lattice and the Canonical Class

Let $\mathrm{Div}_E(\tilde{X})$ denote the lattice of divisors supported entirely on the exceptional locus E. Any element $D \in \mathrm{Div}_E(\tilde{X})$ is an integer linear combination written as a cycle:

$$D = \sum_{i=1}^{n} a_i E_i, \quad a_i \in \mathbb{Z}$$

The intersection product defines a symmetric bilinear form on this lattice:

$$\mathrm{Div}_E(\tilde{X}) \times \mathrm{Div}_E(\tilde{X}) \longrightarrow \mathbb{Z}, \quad (D_1, D_2) \longmapsto D_1 \cdot D_2$$

Definition 1.3 (Adjunction Formula and Numerical Canonical Divisor). Let $K_{\tilde{X}}$ be the canonical divisor of the smooth surface $\tilde{X}$, representing the isomorphism class of the invertible sheaf of holomorphic 2-forms $\Omega^2_{\tilde{X}}$. For each irreducible exceptional component E_i, its geometric genus $g(E_i)$ is related to its self-intersection via the classical adjunction formula:

$$2g(E_i) - 2 = E_i \cdot (E_i + K_{\tilde{X}})$$

Since the matrix $M = (E_i \cdot E_j)$ is non-singular by negative-definiteness, there exists a unique **numerical canonical cycle** $K \in \mathrm{Div}_E(\tilde{X}) \otimes \mathbb{Q}$ uniquely determined by the system of linear adjunction equations:

$$K \cdot E_i = 2g(E_i) - 2 - E_i^2 \quad \text{for all } i = 1, \ldots, n$$

1.2. Combinatorial Translation: The Resolution Graph

The analytic and topological data of the resolution is completely encoded in a weighted dual graph Γ, known as the **resolution graph** or plumbing graph:

- **Vertices:** Each vertex $v_i \in V(\Gamma)$ corresponds uniquely to an irreducible exceptional component E_i. Each vertex is weighted by two integers: its self-intersection number $E_i^2 = -b_i$ (where $b_i \geq 1$) and its genus $g_i = g(E_i)$. If $g_i = 0$, the genus weight is conventionally omitted.
- **Edges:** An edge connects vertex v_i and v_j if $E_i \cap E_j \neq \emptyset$. The number of edges equals the intersection multiplicity $E_i \cdot E_j$.

This graph Γ acts as a plumbing instruction manual for reconstructing the smooth 4-manifold boundary neighborhood $\partial N(E)$, establishing the topological bridge to contact geometry.

2. The Fundamental Cycle (Z) and Artin's Construction Algorithm

In this section, we move from the static topology of the exceptional locus $E = \bigcup E_i$ to its minimal algebraic representative: the **Fundamental Cycle**. In algebraic geometry, we are used to looking at individual curves. However, when studying singularities, we must bundle these curves together into a single non-reduced, effective divisor that acts as a proxy for the structural functions near the singular point.

Let $\mathrm{Div}_E(\tilde{X}) \cong \mathbb{Z}^n$ be the lattice of divisors supported on E. We say a cycle $Y = \sum_{i=1}^n a_i E_i$ is **effective** (written $Y > 0$) if $a_i \geq 0$ for all i, and at least one $a_i > 0$. We write $Y \geq Y'$ if $Y - Y'$ is effective.

2.1. Formal Definition and Intuition

Definition 2.1 (Anti-Nef Cycle)**.** An effective cycle $Y \in \mathrm{Div}_E(\tilde{X})$ is said to be **anti-nef** relative to the exceptional locus if it satisfies the **Artin Condition**:

$$Y \cdot E_i \leq 0 \quad \text{for all } i = 1, 2, \ldots, n$$

Theorem 2.2 (Existence and Uniqueness of the Fundamental Cycle)**.** *Let $M = (E_i \cdot E_j)$ be the negative-definite intersection matrix of a good resolution. There exists a unique, minimal effective cycle $Z \in Div_E(\tilde{X})$, called the **Fundamental Cycle**, such that:*

1. *$Z \cdot E_i \leq 0$ for all $i = 1, \ldots, n$.*
2. *If $Y > 0$ is any other effective cycle satisfying $Y \cdot E_i \leq 0$ for all i, then $Y \geq Z$ (component-wise coefficient domination).*

Note: Why does $Z \cdot E_i \leq 0$ matter? If you think of Z as defining a small neighborhood or a "level set" of a analytic function pulling back to the resolution, functions must vanish along the exception locus. The intersection condition $Z \cdot E_i \leq 0$ guarantees that Z pulls back like a principal divisor of a function vanishing at the origin, meaning it can "pull" the curves inward during a contraction.

2.2. Artin's Construction Algorithm

While the theorem guarantees uniqueness, Michael Artin provided a constructive, algorithmic recipe to explicitly compute Z directly from the resolution graph or intersection matrix.

The Construction Steps

Step 1 (Initialization): Start with the minimal possible nontrivial effective support. Define the initial cycle Z_0 by setting all coefficients to 1:

$$Z_0 = \sum_{i=1}^{n} E_i$$

Set the iteration counter $k = 0$.

Step 2 (The Stopping Test): Calculate the intersection numbers $Z_k \cdot E_i$ for all $i = 1, \dots, n$.

- If $Z_k \cdot E_i \leq 0$ for **all** i, stop. The algorithm has converged, and the fundamental cycle is given by $Z = Z_k$.
- If there exists **at least one** index j such that $Z_k \cdot E_j > 0$, proceed to Step 3.

Step 3 (The Correction Step): Choose any component E_j where the stopping test failed ($Z_k \cdot E_j > 0$). Construct the next cycle by incrementing its coefficient by 1:

$$Z_{k+1} = Z_k + E_j$$

Increment the counter $k \leftarrow k + 1$, and return to Step 2.

Proposition 2.3 (Finiteness and Convergence)**.** *Because the intersection matrix M is symmetric and strictly negative-definite, Artin's algorithm is guaranteed to terminate in a finite number of steps, and the terminal cycle is completely independent of the order in which you choose the violating components E_j in Step 3.*

2.3. A Concrete Example: The A_2 Singularity

Let's compute the fundamental cycle explicitly to see the algorithm in action. Consider the A_2 surface singularity (topologically, this corresponds to the plumbing of two (-2)-curves meeting at a single point).

Its intersection matrix is:

$$M = \begin{pmatrix} E_1 \cdot E_1 & E_1 \cdot E_2 \\ E_2 \cdot E_1 & E_2 \cdot E_2 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 1 & -2 \end{pmatrix}$$

- **Iteration 0:** We initialize $Z_0 = E_1 + E_2$. Now we run the stopping test:

$$Z_0 \cdot E_1 = (E_1 + E_2) \cdot E_1 = E_1^2 + E_2 \cdot E_1 = -2 + 1 = -1 \leq 0 \quad (\checkmark)$$

$$Z_0 \cdot E_2 = (E_1 + E_2) \cdot E_2 = E_1 \cdot E_2 + E_2^2 = 1 - 2 = -1 \leq 0 \quad (\checkmark)$$

  Since both intersection numbers are ≤ 0, the algorithm stops immediately at Step 2.

- **Conclusion:** For the A_2 singularity, the fundamental cycle is $Z = E_1 + E_2$.

3. Artin's Characterization Theorem and Rational Singularities

In classical algebraic geometry, a smooth variety is called rational if it is birationally equivalent to projective space. For singular points, however, **rationality** measures how well-behaved the functions on a resolution are compared to functions on a smooth surface. This section details Michael Artin's breakthrough showing that this analytic behavior is completely determined by the arithmetic genus of the fundamental cycle Z.

3.1. Analytic Definition of Rationality

Let (X, x_0) be a normal complex surface singularity, and let $\pi : \tilde{X} \to X$ be a good resolution. Let $\mathcal{O}_{\tilde{X}}$ be the structure sheaf of holomorphic functions on the smooth resolution surface.

Definition 3.1 (Rational Singularity)**.** The surface singularity (X, x_0) is called a **rational singularity** if the first higher direct image sheaf of the structure sheaf vanishes at the singular point:

$$(R^1\pi_*\mathcal{O}_{\tilde{X}})_{x_0} = 0$$

Equivalently, for a sufficiently small Stein neighborhood $U \subset X$ around x_0, this corresponds to the vanishing of the first cohomology group on the resolution:

$$H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$$

Note: What does $H^1(\tilde{X}, \mathcal{O}_{\tilde{X}}) = 0$ actually mean? By the Riemann-Roch theorem, this cohomology group acts as an obstruction space. When it vanishes, it means there are *plenty* of holomorphic functions available on $\tilde{X}$ that can be pushed down to X. In a very real sense, a rational singularity is the "closest" an isolated singular point can get to behaving like a completely smooth point analytically.

3.2. Artin's Criterion: The Combinatorial Shortcut

Computing the sheaf cohomology group $H^1(\tilde{X}, \mathcal{O}_{\tilde{X}})$ directly requires tracking analytic data over charts. In 1966, Michael Artin proved that you can bypass the sheaf theory entirely by using the fundamental cycle Z computed via the intersection matrix.

Theorem 3.2 (Artin's Characterization Theorem)**.** *Let (X, x_0) be a normal surface singularity, and let Z be its fundamental cycle on a good resolution $\tilde{X}$. The singularity is rational if and only if the **arithmetic genus** of the fundamental cycle is zero:*

$$p_a(Z) = 0$$

3.3. The Arithmetic Genus Formula

To use Artin's theorem, we must define the arithmetic genus of a non-reduced divisor. For any effective cycle $Y = \sum a_i E_i$, its arithmetic genus $p_a(Y)$ is defined via the classical Adjunction Formula on a surface:

$$p_a(Y) = 1 + \frac{1}{2}\left(Y^2 + Y \cdot K_{\tilde{X}}\right)$$

Where $K_{\tilde{X}}$ is the canonical divisor of the resolution surface.

Computing $p_a(Z)$ Purely From the Resolution Graph. We often get stuck because $K_{\tilde{X}}$ is not explicitly given by the graph. However, we can eliminate $K_{\tilde{X}}$ using the individual formulas for the exceptional curves. For each smooth component E_i, we know its geometric genus is $g(E_i)$. Applying adjunction to E_i gives:

$$2g(E_i) - 2 = E_i^2 + E_i \cdot K_{\tilde{X}} \implies E_i \cdot K_{\tilde{X}} = 2g(E_i) - 2 - E_i^2$$

Since $Z = \sum_{i=1}^n c_i E_i$, we can expand the linear term $Z \cdot K_{\tilde{X}}$ directly:

$$Z \cdot K_{\tilde{X}} = \sum_{i=1}^n c_i \left(E_i \cdot K_{\tilde{X}}\right) = \sum_{i=1}^n c_i \left(2g(E_i) - 2 - E_i^2\right)$$

Substituting this back into the main genus formula gives a calculation tool dependent only on the graph data:

$$p_a(Z) = 1 + \frac{1}{2}Z^2 + \frac{1}{2}\sum_{i=1}^{n} c_i \left(2g(E_i) - 2 - E_i^2\right)$$

3.4. A Step-by-Step Rationality Check

Let's test the A_2 singularity from Section 2. We already computed its fundamental cycle as $Z = E_1 + E_2$, where both curves are smooth rational curves ($g(E_1) = g(E_2) = 0$) with self-intersections $E_1^2 = E_2^2 = -2$, and $E_1 \cdot E_2 = 1$.

1. **Compute Z^2:**

$$Z^2 = (E_1 + E_2) \cdot (E_1 + E_2) = E_1^2 + 2(E_1 \cdot E_2) + E_2^2 = -2 + 2(1) - 2 = -2$$

2. **Compute the canonical pairings $E_i \cdot K_{\tilde{X}}$:**

$$E_1 \cdot K_{\tilde{X}} = 2(0) - 2 - (-2) = 0$$

$$E_2 \cdot K_{\tilde{X}} = 2(0) - 2 - (-2) = 0$$

3. **Compute $Z \cdot K_{\tilde{X}}$:**

$$Z \cdot K_{\tilde{X}} = 1(E_1 \cdot K_{\tilde{X}}) + 1(E_2 \cdot K_{\tilde{X}}) = 1(0) + 1(0) = 0$$

4. **Plug into the Arithmetic Genus Formula:**

$$p_a(Z) = 1 + \frac{1}{2}\left(Z^2 + Z \cdot K_{\tilde{X}}\right) = 1 + \frac{1}{2}(-2 + 0) = 1 - 1 = 0$$

Since $p_a(Z) = 0$, Artin's Criterion guarantees that the A_2 singularity is a **rational singularity**. All Du Val/ADE singularities pass this test.

3.5. Properties of Rational Singularities

If a singularity passes Artin's test ($p_a(Z) = 0$), it inherits powerful structural properties:

- **Sub-cycle Genus Bounds:** For any effective sub-cycle $0 < Y \leq Z$, it must hold that $p_a(Y) \leq 0$.
- **Multiplicity and Embedding Dimension:** The multiplicity of the local ring is given simply by $-Z^2$, and the minimal number of generators needed to embed the singularity into complex affine space $\mathbb{C}^N$ is given by $-Z^2 + 1$.

4. Reduced Fundamental Cycles and the Connection to Plamanevskaya–Starkston

We now connect Artin's algebraic framework directly to the geometric mechanics of the Plamanevskaya–Starkston paper (*Unexpected Stein fillings, rational surface singularities, and plane curve arrangements*). The central link hinges on a highly restrictive topological condition: the fundamental cycle being **reduced**.

4.1. Definition: Reduced Fundamental Cycles

A singularity is said to have a **reduced fundamental cycle** if every single coefficient in its fundamental cycle $Z = \sum_{i=1}^{n} c_i E_i$ is identically equal to one:

$$Z = \sum_{i=1}^{n} E_i$$

This is an exceptionally strong property. It means that the unique, minimal anti-nef divisor consists of each exceptional curve counted with multiplicity exactly one.

4.2. The Combinatorial Effect: Graph Valence Bounds

When $Z = \sum E_i$, Artin's condition ($Z \cdot E_i \leq 0$) simplifies down to a purely combinatorial constraint on the resolution (or plumbing) graph. For any given vertex E_i in the graph, we can split its intersection product into its self-intersection and its adjacent neighbors:

$$Z \cdot E_i = \left(\sum_{j=1}^{n} E_j \right) \cdot E_i = E_i^2 + \sum_{j \neq i} (E_j \cdot E_i) \leq 0$$

In a good resolution graph, $\sum_{j \neq i}(E_j \cdot E_i)$ is exactly the **valence** of the vertex E_i (the number of edges connected to it). Therefore, a rational singularity has a reduced fundamental cycle if and only if every vertex in the graph satisfies:

$$-E_i^2 \geq \text{valence}(E_i)$$

This inequality acts as a strict structural filter. It rules out configurations with high branching or loosely constrained negative weights.

5. Task: Verification

The task is as follows:

1. Take the A_k resolution graph (a linear chain of vertices with weights $= -2$).
2. Set up the system of inequalities ($\sum_{i=1}^{k} a_i E_i \cdot E_j \leq 0 \quad \forall j$)
3. Show that $a_i = 1 \forall i$ is the unique minimal solution. Ultimately, this shows that singularities have reduced fundamental cycles.

The A_k Resolution Graph

The A_k singularity resolution consists of a linear chain of k smooth rational exceptional curves, $E_1, E_2, \ldots, E_k$. Its topological data is explicitly defined by the following intersection configurations:

- $E_i \cdot E_i = -2$ for all $i = 1, \ldots, k$.
- $E_i \cdot E_{i+1} = 1$ for all $i = 1, \ldots, k-1$.
- $E_i \cdot E_j = 0$ for all $|i - j| > 1$.

The Fundamental Cycle

The intersection matrix $M = (E_i \cdot E_j)$ for the A_k singularity is a symmetric, tridiagonal matrix of size $k \times k$. It is completely described by:

$$M_{i,j} = \begin{cases} -2 & \text{if } i = j \\ 1 & \text{if } |i-j| = 1 \\ 0 & \text{otherwise} \end{cases}$$

Following Artin's initialization, we define our initial test cycle Z_0 by setting all coefficients $a_i = 1$:

$$Z_0 = \sum_{i=1}^{k} E_i$$

We define our initial test cycle Z_0 by setting all coefficients $a_i = 1$:

$$Z_0 = \sum_{i=1}^{k} E_i = E_1 + E_2 + \cdots + E_k$$

For the left endpoint curve E_1, the intersection product expands as:

$$Z_0 \cdot E_1 = (E_1 + E_2 + E_3 + \cdots + E_k) \cdot E_1$$
$$Z_0 \cdot E_1 = E_1^2 + E_2 \cdot E_1 + 0 + \cdots + 0$$

Plugging in the weights $E_1^2 = -2$ and $E_2 \cdot E_1 = 1$:

$$Z_0 \cdot E_1 = -2 + 1 = -1 \leq 0 \quad (\checkmark)$$

For any general interior curve E_j where $1 < j < k$, the product expands as:

$$Z_0 \cdot E_j = (E_1 + \cdots + E_{j-1} + E_j + E_{j+1} + \cdots + E_k) \cdot E_j$$
$$Z_0 \cdot E_j = E_{j-1} \cdot E_j + E_j^2 + E_{j+1} \cdot E_j$$

Plugging in the weights $E_j^2 = -2$ and the neighbor intersections 1:

$$Z_0 \cdot E_j = 1 - 2 + 1 = 0 \leq 0 \quad (\checkmark)$$

For the right endpoint curve E_k, the intersection product expands as:

$$Z_0 \cdot E_k = (E_1 + \cdots + E_{k-1} + E_k) \cdot E_k$$
$$Z_0 \cdot E_k = E_{k-1} \cdot E_k + E_k^2$$

Plugging in the weights $E_k^2 = -2$ and $E_{k-1} \cdot E_k = 1$:

$$Z_0 \cdot E_k = 1 - 2 = -1 \leq 0 \quad (\checkmark)$$

We have verified that the initial effective cycle $Z_0 = \sum_{i=1}^{k} E_i$ satisfies $Z_0 \cdot E_j \leq 0$ for all $j = 1, \ldots, k$. By Artin's uniqueness theorem, if Y is any effective cycle ($Y > 0$) such that $Y \cdot E_j \leq 0$ for all j, then $Y \geq Z_0$ component-wise.

Because our initial guess Z_0 achieves the minimal possible positive integer coefficients ($a_i = 1$ for all i), it is impossible to subtract any further components without violating the condition that the fundamental cycle must be strictly effective ($Z > 0$). Therefore, $Z = Z_0$ is the unique minimal solution.

Conclusion: Because $Z = \sum_{i=1}^{k} E_i$, all coefficients are identically 1. This formally completes the verification that the A_k surface singularities have strictly **reduced** fundamental cycles.

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